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Rules Contributing to Suspicious AWS EC2 Activity Alert

The following rules are used to identify suspicious activity within AWS EC2 logs. Any one or more of these will trigger Suspicious AWS EC2 Activity Alert. Details for each rule can be viewed by clicking the **More Details** link in the description.

Title	Description

AWS EC2 Startup Shell Script Change	Detects changes to the EC2 instance startup script. The shell script will be executed as root/SYSTEM every time the specific instances are booted up.
	
	More details
	
	Rule ID
	
	`aws_8`
	
	Query
	
	`{'selection_source': {'eventSource': 'ec2.amazonaws.com', 'requestParameters_attribute': 'userData', 'eventName': 'ModifyInstanceAttribute'}, 'condition': 'selection_source'}`
	
	Log Source
	
	Stellar Cyber AWS configured for:
	
	• AWS Cloudtrail
	
	Rule Source
	
	SigmaHQ,1ab3c5ed-5baf-417b-bb6b-78ca33f6c3df
	
	Author: faloker
	
	Tactics, Techniques, and Procedures
	
	TA0002, T1059
	
	References
	
	• https://github.com/RhinoSecurityLabs/pacu/blob/866376cd711666c775bbfcde0524c817f2c5b181/pacu/modules/ec2__startup_shell_script/main.py#L9
	
	Severity
	
	75

Title	Description
	Suppression Logic Based On • service_id • aws.eventSource • aws.eventName • stellar.rule_id **Additional Information** **Maturity**: experimental **Creation Date**: 2020/02/12 **Risk Level**: high **False Positives** • Valid changes to the startup script

AWS EC2 Network Access Control List Creation	Identifies the creation of an AWS Elastic Compute Cloud (EC2) network access control list (ACL) or an entry in a network ACL with a specified rule number.
	
	More details
	
	Rule ID
	`aws_30`
	
	Query
	`{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': ['CreateNetworkAcl', 'CreateNetworkAclEntry']}, 'condition': 'selection1 and selection2'}`
	
	Log Source
	Stellar Cyber AWS configured.
	
	Rule Source
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	TA0003, T1133
	
	References
	N/A
	
	Severity
	25
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• stellar.rule_id
	
	Additional Information
	Maturity: production
	Creation Date: 2020/06/04
	Risk Level: low
	False Positives
	• Network ACL's may be created by a network administrator. Verify whether the user identity, user agent, and/or hostname should be making changes in your environment. Network ACL creations by unfamiliar users or hosts should be investigated. If known behavior is causing false positives, it can be exempted from the rule.

AWS EC2 Network Access Control List Deletion	Identifies the deletion of an Amazon Elastic Compute Cloud (EC2) network access control list (ACL) or one of its ingress/egress entries. More details **Rule ID** `aws_45`

Title	Description
	Query `{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': ['DeleteNetworkAcl', 'DeleteNetworkAclEntry']}, 'condition': 'selection1 and selection2'}` **Log Source** Stellar Cyber AWS configured. **Rule Source** Developed internally by Stellar Cyber **Tactics, Techniques, and Procedures** TA0005, T1562 **References** N/A **Severity** 50 **Suppression Logic Based On** • service_id • aws.eventSource • aws.eventName • stellar.rule_id **Additional Information** **Maturity**: production **Creation Date**: 2020/05/26 **Risk Level**: medium **False Positives** • Network ACL's may be deleted by a network administrator. Verify whether the user identity, user agent, and/or hostname should be making changes in your environment. Network ACL deletions by unfamiliar users or hosts should be investigated. If known behavior is causing false positives, it can be exempted from the rule.

AWS EC2 Snapshot Activity	An attempt was made to modify AWS EC2 snapshot attributes. Snapshots are sometimes shared by threat actors in order to exfiltrate bulk data from an EC2 fleet. If the permissions were modified, verify the snapshot was not shared with an unauthorized or unexpected AWS account.
	More details
	Rule ID
	`aws_51`
	Query
	`{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': 'ModifySnapshotAttribute'}, 'condition': 'selection1 and selection2'}`
	Log Source
	Stellar Cyber AWS configured.
	Rule Source
	Developed internally by Stellar Cyber
	Tactics, Techniques, and Procedures
	TA0010, T1537

Title	Description

Refere
N/A

Severity

50

Suppression Logic Based On

- service_id
- aws.eventSource
- aws.eventName
- stellar.rule_id

Additional Information

Maturity	Creation Date	Risk Level	False Positives
production	2020/06/24	medium	• IAM users may occasionally share EC2 snapshots with another AWS account belonging to the same organization. If known behavior is causing false positives, it can be exempted from the rule.

AWS EC2 VM Export Failure

Identifies an attempt to export an AWS EC2 instance. A virtual machine (VM) export may indicate an attempt to extract or exfiltrate information.

More details

Rule ID

`aws_54`

Query

```
{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': 'CreateInstanceExportTask'}, 'condition': 'selection1 and selection2'}
```

Log Source

Stellar Cyber AWS configured.

Rule Source

Developed internally by Stellar Cyber

Tactics, Techniques, and Procedures

TA0009, T1005, TA0010, T1537

References
N/A

Severity

25

Suppression Logic Based On

- service_id
- aws.eventSource
- aws.eventName
- stellar.rule_id

Additional Information

Maturity	Creation Date	Risk Level	False Positives
production	2021/04/22	low	• VM exports may be done by a system or network administrator. Verify whether the user identity, user agent, and/or hostname should be making changes in your environment. VM exports from

Title	Description
	Maturity / **Creation Date** / **Risk Level** / **False Positives** unfamiliar users or hosts should be investigated. If known behavior is causing false positives, it can be exempted from the rule.

AWS EC2 Full Network Packet Capture Detected	Identifies potential Traffic Mirroring in an Amazon Elastic Compute Cloud (EC2) instance. Traffic Mirroring is an Amazon VPC feature that you can use to copy network traffic from an Elastic network interface. This feature can potentially be abused to exfiltrate sensitive data from unencrypted internal traffic.
	
	More details
	
	Rule ID
	`aws_57`
	
	Query
	`{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': ['CreateTrafficMirrorFilter', 'CreateTrafficMirrorFilterRule', 'CreateTrafficMirrorSession', 'CreateTrafficMirrorTarget']}, 'condition': 'selection1 and selection2'}`
	
	Log Source
	Stellar Cyber AWS configured.
	
	Rule Source
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	TA0009, T1074, TA0010, T1020
	
	References
	N/A
	
	Severity
	50
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• stellar.rule_id
	
	Additional Information
	Maturity / **Creation Date** / **Risk Level** / **False Positives**
	production / 2021/05/05 / medium / • Traffic Mirroring may be done by a system or network administrator. Verify whether the user identity, user agent, and/or hostname should be making changes in your environment. Traffic Mirroring from unfamiliar users or hosts should be investigated. If known behavior is causing false positives, it can be exempted from the rule.

AWS EC2 Encryption Disabled	Identifies disabling of Amazon Elastic Block Store (EBS) encryption by default in the current region. Disabling encryption by default does not change the encryption status of your existing volumes. More details **Rule ID**

Title	Description
	aws_5  **Query** `{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': 'DisableEbsEncryptionByDefault'}, 'condition': 'selection1 and selection2'}` **Log Source** Stellar Cyber AWS configured. **Rule Source** Developed internally by Stellar Cyber **Tactics, Techniques, and Procedures** TA0040, T1565 **References** N/A **Severity** 50 **Suppression Logic Based On** • service_id • aws.eventSource • aws.eventName • stellar.rule_id **Additional Information** **Maturity**: production **Creation Date**: 2020/06/05 **Risk Level**: medium **False Positives** • Disabling encryption may be done by a system or network administrator. Verify whether the user identity, user agent, and/or hostname should be making changes in your environment. Disabling encryption by unfamiliar users or hosts should be investigated. If known behavior is causing false positives, it can be exempted from the rule.

EC2 Snapshot Attribute Modification	The following analytic utilizes AWS CloudTrail events to identify when an EC2 snapshot permissions are modified.
	More details
	Rule ID
	`aws_76`
	Query
	`{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': 'ModifySnapshotAttribute'}, 'condition': 'selection1 and selection2'}`
	Log Source
	Stellar Cyber AWS configured.
	Rule Source
	Developed internally by Stellar Cyber
	Tactics, Techniques, and Procedures
	TA0010, T1537,

Title	Description
	References • https://labs.nettitude.com/blog/how-to-exfiltrate-aws-ec2-data/ • https://stratus-red-team.cloud/attack-techniques/AWS/aws.exfiltration.ec2-share-ebs-snapshot/ • https://hackingthe.cloud/aws/enumeration/loot_public_ebs_snapshots/ **Severity** 60 **Suppression Logic Based On** • service_id • aws.eventSource • aws.eventName • stellar.rule_id **Additional Information** **Maturity**: production **Creation Date**: 2023-03-20 **Risk Level**: medium **False Positives**: • It is possible that an AWS admin has legitimately modified permissions of EC2 Snapshot.

AWS EC2 Security Group Deleted	An EC2 security group has been deleted.
	More details
	Rule ID
	`aws_77`
	Query
	`{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': 'DeleteSecurityGroup'}, 'selection3': {'userAgent': 'cloudformation.amazonaws.com'}, 'condition': 'selection1 and selection2 and not selection3'}`
	Log Source
	Stellar Cyber AWS configured.
	Rule Source
	Developed internally by Stellar Cyber
	Tactics, Techniques, and Procedures
	TA0003, T1098
	References
	N/A
	Severity
	50
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• stellar.rule_id
	Additional Information
	Maturity: test
	Creation Date: 2022/01/05
	Risk Level: medium
	False Positives: N/A

Title	Description

AWS EC2 Security Group Modified	An EC2 security group has been modified.
	
	More details
	
	Rule ID
	`aws_78`
	
	Query
	`{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': 'DeleteRouteTable'}, 'selection3': {'eventName': 'DeleteSubnet'}, 'selection4': {'eventName': 'CreateDBSubnetGroup'}, 'selection5': {'eventName': 'DeleteDBSubnetGroup'}, 'selection6': {'eventName': 'ModifyDBSubnetGroup'}, 'selection7': {'userAgent': 'cloudformation.amazonaws.com'}, 'condition': 'selection1 and (selection2 or selection3 or selection4 or selection5 or selection6) and not selection7'}`
	
	Log Source
	Stellar Cyber AWS configured.
	
	Rule Source
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	TA0003, T1098
	
	References
	N/A
	
	Severity
	50
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• stellar.rule_id
	
	Additional Information
	Maturity: test
	Creation Date: 2022/01/05
	Risk Level: medium
	False Positives: N/A

AWS EC2 Security Group Created	An EC2 security group has been created.
	
	More details
	
	Rule ID
	`aws_79`
	
	Query
	`{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': 'CreateSecurityGroup'}, 'selection3': {'userAgent': 'cloudformation.amazonaws.com'}, 'condition': 'selection1 and selection2 and not selection3'}`
	
	Log Source
	Stellar Cyber AWS configured.
	
	Rule Source

Title	Description
	Develo nternally by Stellar Cyber **Tactics, Techniques, and Procedures** TA0003, T1098 **References** N/A **Severity** 50 **Suppression Logic Based On** • service_id • aws.eventSource • aws.eventName • stellar.rule_id **Additional Information** **Maturity**: test **Creation Date**: 2022/01/05 **Risk Level**: medium **False Positives**: N/A

AWS Credential Access GetPasswordData	This detection analytic identifies GetPasswordData API call made to your AWS account. Attackers can retrieve the encrypted administrator password for a running Windows instance.
	
	More details
	
	Rule ID
	
	`aws_88`
	
	Query
	
	`{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': 'GetPasswordData'}, 'condition': 'selection1 and selection2'}`
	
	Log Source
	
	Stellar Cyber AWS configured.
	
	Rule Source
	
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	
	TA0006, T1110.001
	
	References
	
	• https://stratus-red-team.cloud/attack-techniques/AWS/aws.credential-access.ec2-get-password-data/
	
	Severity
	
	70
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• stellar.rule_id
	
	Additional Information
	
	Maturity: production
	Creation Date: 2022-08-10
	Risk Level: medium
	False Positives:
	• Administrator tooling or automated scripts may make these calls

Title	Description

AWS VPC Network ACL Modified	The ACL for a VPC has been modified.
	
	More details
	
	Rule ID
	`aws_104`
	
	Query
	`{'selection1': {'eventSource': 'ec2.amazonaws.com'}, 'selection2': {'eventName': 'CreateACLEntry'}, 'selection3': {'eventName': 'DeleteACL'}, 'selection4': {'eventName': 'DeleteACLEntry'}, 'selection5': {'eventName': 'UpdateACLAssociation'}, 'selection6': {'userAgent': 'cloudformation.amazonaws.com'}, 'condition': 'selection1 and (selection2 or selection3 or selection4 or selection5) and not selection6'}`
	
	Log Source
	Stellar Cyber AWS configured.
	
	Rule Source
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	TA0003, T1098
	
	References
	N/A
	
	Severity
	50
	
	Suppression Logic Based On
	• service_id
	• aws.eventSource
	• aws.eventName
	• stellar.rule_id
	
	Additional Information
	Maturity: test
	Creation Date: 2022/01/05
	Risk Level: medium
	False Positives: N/A

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